Baseline Results: PIM-malloc Reproduction and Maintained Controls

CS5290 Project Group

I. Original artifact baseline

These results reproduce the paper's allocation microbenchmark using the preserved released runtime, not the repaired contribution runtime. The reported paper-compatible aggregation is batch completion time divided by 128 requests/tasklet. Maximum absolute deviation from a printed label is 7.20%. The original WRAM layout permits 256 KiB and therefore does not demonstrate physical 64 KiB fit.

TABLE I
Original Figure 15 artifact reproduction: batch time divided by 128 requests per tasklet, in microseconds.

Tasklets	Bytes	Paper buddy	Measured buddy	Paper SW	Measured SW	Buddy/SW
1	32	630	584.66	6.9	6.99	83.66
1	256	520	492.92	32	32.12	15.35
1	4096	350	362.71	310	312.71	1.16
16	32	13000	13026.39	51	51.86	251.17
16	256	11000	10911.77	430	457.72	23.84
16	4096	7700	7810.92	6700	6808.97	1.15

The original simulator layout is preserved here. These latencies are separate from the repaired allocator and 64 KiB WRAM configuration used for contributions.

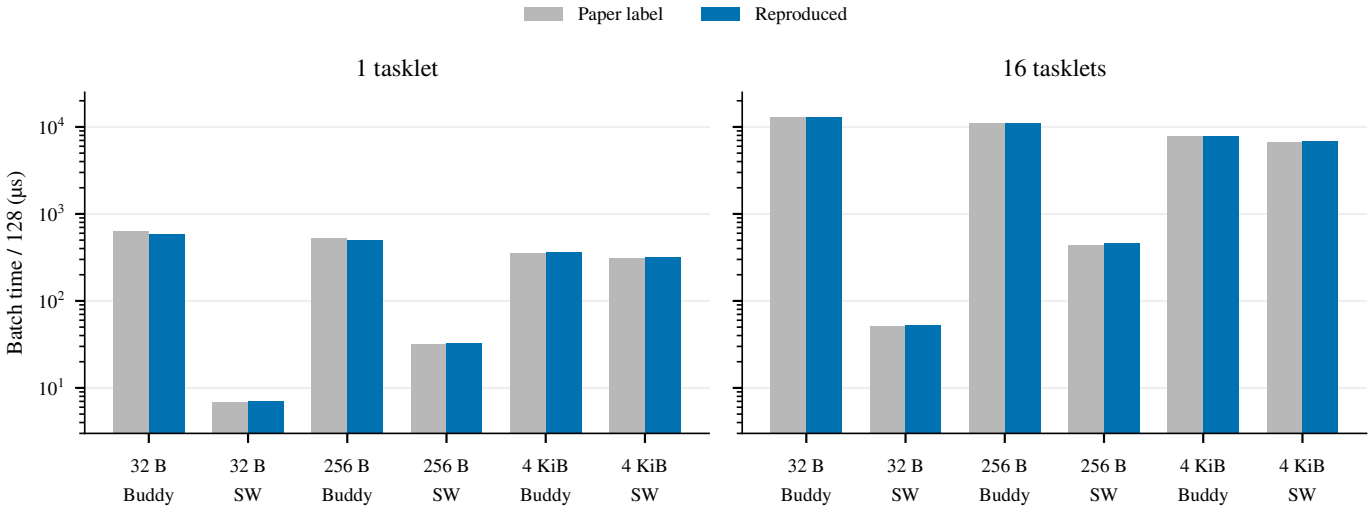

Fig. 1. Original paper labels and reproduced allocator latencies.

II. Maintained baseline used for contributions

Free/bitmap routing and empty-page reclamation are repaired identically in all maintained variants. The contribution runtime caps WRAM addresses at 65,536 B. The final baseline is scan fixed-2, which limits only completely empty retained pages/class. It uses a 32 MiB payload heap and 256 KiB empty-byte budget/tasklet. Its raw times are the B columns below; the adjacent contribution columns provide context. These values must not be merged with the original artifact as though only one optimization changed.

III. Available data

The accompanying tables include the original six paper/measured pairs, the final baseline conditions and exact phase/traffic definitions. All contribution speedups use the matching stage control. Simulated frequency is 350 MHz. Graphs are reduced ports and KV is allocation-only.

TABLE II
FINAL WORK RUNTIME (MILLISECONDS) FOR EVERY STAGE ON FIVE COMPLEMENTARY CASES.

Case	B	IA	IF	SA	SF	PA	PF
Larson 8–4095 B (4T)	160.089	164.537	159.826	24.026	19.894	19.172	16.255
CloudPhysics cache (4T)	122.773	122.925	122.784	114.586	114.528	113.023	113.075
Larson 8–8191 B (4T)	320.457	342.572	331.666	216.369	232.613	204.807	188.202
Gowalla churn (4T)	148.847	26.172	24.961	26.918	25.830	26.954	25.827
PrIM selection (4T)	14.134	14.180	14.180	14.135	14.135	14.133	14.133

B: scan fixed-2; I: indexed; S: indexed plus 64 KiB slabs; P: indexed plus slabs plus pinned metadata. A uses adaptive interval 64; F retains at most two empty spans/class. All rows use four tasklets.

TABLE III
FINAL PAPER-DERIVED CASES: THE BASELINE AND BOTH FULL-STACK POLICIES.

Case	B work (ms)	PA work (ms)	PF work (ms)	PA speedup	PF speedup	PF total speedup
Gowalla linked blocks (4T)	19.330	15.479	15.198	1.249	1.272	1.335
Gowalla growing arrays (4T)	5.752	5.845	5.494	0.984	1.047	1.070
KV growth replay (4T)	24.382	19.657	19.625	1.240	1.242	1.210
32 B microbenchmark (1T)	0.875	1.077	0.990	0.813	0.884	0.799
32 B microbenchmark (16T)	6.938	5.719	5.648	1.213	1.228	1.105
256 B microbenchmark (1T)	2.076	2.005	1.925	1.035	1.079	0.994
256 B microbenchmark (16T)	34.201	27.750	27.641	1.232	1.237	1.192
4 KiB microbenchmark (1T)	20.912	1.858	1.778	11.255	11.763	1.512
4 KiB microbenchmark (16T)	508.474	22.335	22.243	22.766	22.860	6.746

Graph cases are reduced ports; KV is an allocation-only replay and excludes attention/serving. Original request sizes and 128 microbenchmark requests/tasklet are retained. Values below one are regressions.